YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 4 OF 10 · 40 MINUTES

The white powder 白色粉末之谜

Testing every sample the same way and recording results in a grid lets us identify an unknown against known samples; a grid can become a key.

- I can test five known powders and the scene powder systematically and record results in a grid.
- I can use my results to identify the scene powder and turn the grid into a flowchart key.

systematic 系统的 · **soluble** 可溶的 · **insoluble** 不溶的 · **fizz** 冒气泡 · **react** 反应 · **starch** 淀粉 · **iodine solution** 碘液 · **crystal** 晶体 · **results grid** 结果记录表 · **identification key** 检索表

THE LESSON

Do Now	4	Recap questions; stick dry Week 3 chromatograms into Week 3 worksheet; goals. Slides 2 to 3
Hook	3	Hold up Evidence 3 bag: which of five canteen powders? Hands-up vote, numbers on board. Slide 4
Learn	7	Three tests and positive results; model one powder; be systematic; goggles on. Slides 5 to 7
Do	18	Test 5 knowns, scene powder last; Recorder fills grid (12 min); key (6 min). Slides 8 to 10
Think and Reflect	8	Name powder; linked suspects; evidence log row 4; confidence vote; exit ticket. Slides 11 to 12

Hands-on: Groups of 4 test five known white powders and the scene powder: magnifier, water, vinegar, iodine (optional).

Collect or assess: Results grid and flowchart key; evidence log row 4, suspect tracker and confidence row; exit ticket; logbook if time.

SAFETY

- Goggles on before vinegar or iodine; iodine stains: teacher or Safety Officer only
- Wheat or corn allergy: no flour or cornflour handling; give them the Recorder role
- Never taste powders or liquids; do not blow or shake powders; wash hands after

WATCH OUT

- Weak fizzing gets missed: watch each well for 10 seconds; one clean spoon per powder.
- Many groups will name Dr Huang: stay neutral, do not confirm or deny; ask how sure they are.

Language: Frame: 'The scene powder matches ___ because it ___.' Soluble vs solvent trap; fizz, starch new. Band 1: buddy, Measurer

Next week: Week 5: print checkpoint papers; laminate colour pH chart; red cabbage indicator day before (25 min, chill).

PREP FOR ONE CLASS

before the lesson about 70 min (35 could be done by someone else); after about 34 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book printing: 24 worksheets (48 sheets), 6 lab cards, kit card, A3 poster 5 min · reprographics
- [ ] Get or buy powders and vinegar; order pots, bags, sticks, card, labels 10 min · teacher

THE WEEK BEFORE

- [ ] Ask prep room for goggles, tiles, cups, droppers, trays, iodine if used 5 min · technician
- [ ] Allergy check; fill and label 6 powder sets, 6 Evidence 3 bags; fizz test 15 min · teacher
- [ ] Laminate and trim 6 powder testing lab cards (A5 pouches) 10 min · technician

THE DAY BEFORE

- [ ] Find Week 3 chromatograms and worksheets; plan roles and seating 10 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Pour vinegar, fill 36 water cups, set out 6 trays and table materials 15 min · technician

AFTER THE LESSON

- [ ] Clean up: empty cups, rinse tiles, wipe tables, wash hands, pack class box 4 min · students
- [ ] Wash kit; store iodine; refill powder pots, bags and vinegar for next class 15 min · technician
- [ ] Read exit tickets and grids; note Week 5 recap and unfinished Stretch keys 10 min · teacher
- [ ] Put up A3 key poster once every class has done Week 4 5 min · teacher
- [ ] Optional: keep unopened Evidence 3 bags, pots, black card for Weeks 7 and 8 optional 2 min · technician

PRINT

- Worksheet x24 (3 pages on 2 A4 sheets; pages 1 and 2 back to back)
- Powder testing lab card x6 (A5, laminated)
- Identification key poster x1 (A3, wall from Week 5)
- Teacher copy: kit card x1 (A5)

GET OR MAKE (NEW THIS LESSON)

- Salt, sugar, cornflour, plain flour: about 12 tsp each
- Baking soda about 30 tsp; white vinegar about 180 ml
- Small lidded pots x30; small zip bags x6
- Stirring sticks x36; spoons or lolly sticks x36
- Black card x6; sticky labels x42
- Laminating pouches, A5 x6
- Safety goggles x24
- Spotting tiles (12 wells) x6
- Small clear plastic cups x42
- Droppers (plastic pipettes) x6
- Iodine solution dropper bottles x6 (optional)
- Teacher's measuring teaspoon
- Paper towels and tap water (lab supply)
- Make: known powder sets x6 (5 labelled pots each)
- Make: Evidence 3 bags x6 (about 3 tsp baking soda)
- Teacher demo set (optional)

Have ready from earlier weeks: Class box: case files, logbooks, Week 3 worksheets and dry chromatograms, Role cards and frame card sets x6, Science Dictionaries and pencils x24, Trays and magnifiers x6; cloths, Clear sticky tape (Weeks 2 and 3)